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Induced Electric Field in Learning-Based Head Models with Smooth Conductivity for Exposure to Uniform Low-Frequency Magnetic Fields

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***Abstract*—Computational human models generated from medical images have been widely used to assess induced electric field for exposure to electromagnetic field. Traditional methods to develop human models include tissue segmentation, which involves huge effort in identifying tissues from medical images. When such models are applied to low-frequency electromagnetic dosimetry, computational artifacts result in substantial error. Deep learning techniques have been utilized to map medical images directly to tissue electrical conductivity, generating human models with smooth transitions in tissue conductivity across tissue boundaries and even within the same tissue. In this study, eight head models with smoothed conductivities were generated using the deep learning network. The induced electric fields in the models were assessed for exposure to a uniform low-frequency magnetic field and were compared with traditional segmented models. Computational results showed that the induced electric field distributions in learning-based and segmented models were consistent, and the former was smoother. The differences in the 99th to 99.99th percentile values between non-uniform and segmented models were within 8% and 13% for gray and white matter, respectively. The staircasing errors were suppressed in the learning-based models because of the smooth transition of the conductivity values, especially at the tissue interface. The inter-subject variation of the maximum electric fields was smaller for the non-uniform models than for the segmented models, with a relative standard deviation within 12% for non-uniform models and 22% for segmented models. This difference is much smaller than the reduction factor of 3 associated with the numerical uncertainty set in the International Commission on Non-Ionizing Radiation Protection 2010 guidelines. Our findings could be helpful in deriving appropriate reduction factor in international guidelines, which is used for setting the limit from the threshold of adverse health effects.**

***Index Terms*—Electromagnetic exposure, electromagnetic safety, deep learning, low frequency, medical images, standardization.**

I. Introduction

THE International Commission on Non-Ionizing Radiation Protection (ICNIRP) guidelines and the IEEE C95.1 standard have been established to protect against central nervous system (CNS) or peripheral nervous system (PNS) tissues stimulation by low-frequency (LF) electromagnetic field exposure (< 100 kHz) [1], [2]. Two types of metrics have been used for protection. The first is basic restriction in the ICNIRP guidelines or the dosimetric reference limit in the IEEE standard, which is defined in terms of the induced electric field strength inside the human body. The limit is derived from the threshold of adverse effects considering the reduction factor [3]. The other metric is reference level in the ICNIRP guidelines or exposure reference level in the IEEE standard, which is defined as the electric field or magnetic field strength in free space without the presence of a human body, for ease of compliance assessment. The reference levels were determined using dosimetric computations, which relate external and induced fields in human models in a conservative manner. For general public exposed to an electromagnetic field above 300 Hz and occupational exposure above 400 Hz, the ICNIRP guidelines primarily protect from PNS tissues stimulation [1], while the IEEE exposure reference level protects against adverse PNS effects at frequencies above ~750 Hz [2]. At frequencies lower than the transition frequencies, the limits were based on the threshold of magnetophosphenes [4] and the brain for protection against CNS tissues stimulation.

In the ICNIRP guidelines (2010), a reduction factor of 3 was considered to set the limit, considering the computational uncertainty. The rationale for this factor has not been explicitly mentioned. If the uncertainty associated with the dosimetry is quantified, it would be useful for setting the limit in a more scientific manner [5]. Thus, continued improvements in computational models and methods are critical to establish the scientific basis for exposure standards, as mentioned in the IEEE ICES research agenda [6] and ICNIRP "Gaps in

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knowledge" document [7].

Beginning two decades ago, many studies assessed the dosimetry quantities using voxel-based anatomical human models for exposures to LF electromagnetic fields. The main purpose of such studies was to derive the relationship between external and internal electric fields to determine the reference level [8]–[19] and to assess the compliance of realistic products [20]–[24]. Such models were generated by segmenting each tissue based on the level of medical image pixels. The corresponding conductivities [25], [26], as reported in some database, were then assigned to each tissue, yielding uniform conductivity values for each tissue.

In numerical dosimetry assessments, one major source of uncertainty is the representation of the tissue conductivities [6], [27]–[30]. Because of the abrupt changes in electric properties across the tissue interface, computed fields inside the voxel models suffer from the well-known staircasing error. Computational artifacts can also originate from segmentation errors in an anatomical model, known as partial volume effect, because the anatomical imaging quality is commonly demonstrated in a millimeter resolution [31]. To resolve this [7], intercomparisons have been conducted [32], [33]. Although the uncertainties related to computational methods can somehow be clarified, uncertainties that originate from tissue conductivity assignment cannot be resolved.

Studies have been conducted to reduce the computational artifacts associated with segmented human models. In [27], the 99th percentile value of the induced electric field was suggested as a practical surrogate for maximum electric field to suppress the effects of computational artifacts. However, there are concerns about using 99th percentile value, especially in the case of localized exposure scenarios [17], [34]. For instance, the skin contains many voxels and are distributed all over the body. Filtering the top 1% electric field values may exclude the real high electric fields. Several techniques including pre- and post-processing methods were proposed. A conductivity-smoothing algorithm before the numerical calculation of induced electric field was proposed in [34]. The 99.9th percentile value for localized exposure was proposed in [35], but this choice is still arbitrary. In another study [36], an outlier removal method based on the frequency distribution of the highest 1% electric field strengths was proposed. A study [37] compared the induced electric field calculated using voxelized and tetrahedral head models for LF magnetic exposures. Results revealed that the maximum electric fields were erroneous for both head models and the 99.9th to 99.99th percentile was suggested to eliminate numerical errors for head tissues.

Mapping the conductivity from MRI has been demonstrated *in vivo* [38], [39]. Deep learning has been recently proposed to automatically compute conductivities [40]–[43]. The U-net architecture was used to estimate brain conductivity using electrical properties tomography [40]. A convolutional neural network called Higher3Dnet was used to estimate the conductivity of the human pelvis [41]. In our previous study, we developed head models with non-uniformly distributed tissue conductivities constructed directly from T1 and T2 images of subjects using a learning network [43]. These models feature smooth transition of the conductivity values across the tissue boundaries and non-uniformly distributed conductivity values even within the same tissue. The models were then adopted in the evaluation of the excitation sites for the transcranial magnetic stimulation of the subject. The computed electric field shows smoother distributions of the induced electric field with lowered staircasing errors inside the human head compared to those for segmented head models. In previous studies [44], [45], such non-uniform head models were also used for assessing radio-frequency exposures. Segmentation-free models provide lower estimation errors than segmented models, even for the EEG source localization [46].

While using learning-based non-uniform conductivity seems to be a promising approach, such models have not been used for numerical analysis for LF exposure standardization. In this study, eight head models were developed using CondNet. The induced electric field was assessed for exposure to uniform LF magnetic fields. Dosimetry quantities in gray matter (GM) and white matter (WM) were analyzed and were compared with those obtained using segmented models. To the best of the authors' knowledge, this is the first study to quantify the effect of the model conductivity smoothness in a head model for LF exposure scenarios.

II. Models and Method

A. Segmented and Non-uniform Head Models

MRIs of eight subjects from the Brian Multimodality Dataset [47] were used to develop voxel-based head models. Firstly, we considered traditional segmented head models. A semi-automatic procedure was used to segment human head tissues from the images [48]. The head models contained 13 different head tissues, including skin, fat, muscle, GM, WM and cerebrospinal fluid (CSF). Their conductivities at 50 Hz were obtained using the 4-Cole-Cole dispersion model and are summarized in Table I [25].

TABLE I
Conductivities of Head Tissues at 50 Hz

Tissue	σ [S/m]	Tissue	σ [S/m]
Blood	0.7	GM	0.08
Bone (canc.)	0.08	Mucous tissue	0.1
Bone (cort.)	0.02	Muscle	0.24
Cerebellum	0.1	Skin	0.1
CSF	2.0	V. Humor	1.5
Dura	0.5	WM	0.05
Fat	0.045		

The second set of head models were developed by deep learning network CondNet [43]. CondNet is a deep learning convolutional neural network that enable the estimation of voxel dielectric properties such as conductivities and permittivity directly from MRI scans without segmentation. The network is designed to map the input anatomical images to conductivity maps of human head. The network is trained using standard data processing pipeline that consider segmentation of

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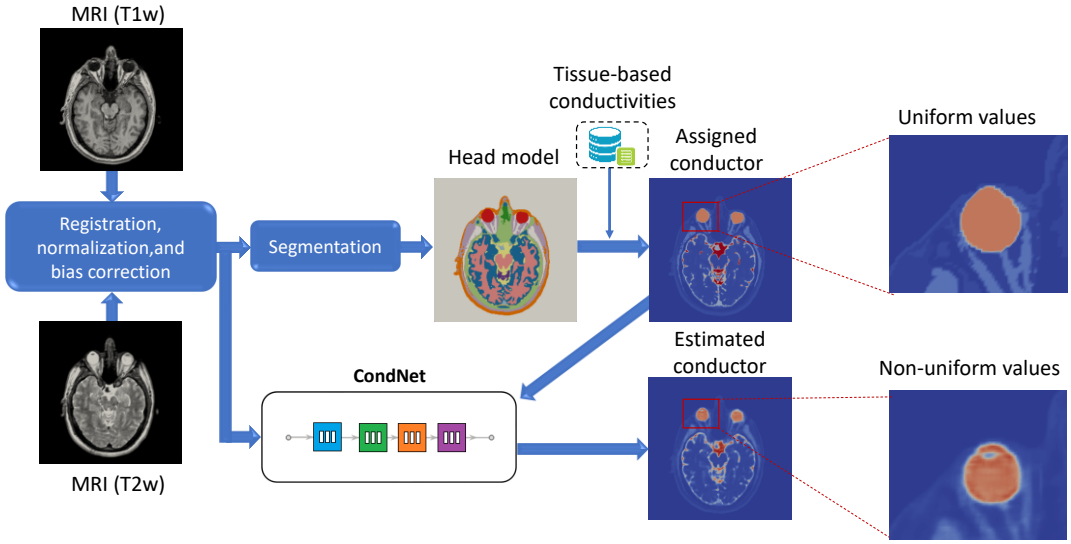

Fig. 1. Pipeline of head model generation based on MRI images using deep learning network CondNet.

several subjects to generate a training dataset, and then new subjects can be developed directly from MRI scans. The whole process of the pipeline is shown in Fig. 1. More technical details can be found in [41], [42].

As shown in Fig. 2, one potential advantage of CondNet is that it can estimate tissue conductivity in a smooth pattern that is highly correlated with the true distribution, where abrupt changes are not common. This method does not require the computational burden of dense segmentation.

B. Exposure Scenario

Uniform magnetic field of 1 mT at 50 Hz was considered. The direction of the magnetic field was chosen to be aligned with the vertical axis as shown in Fig. 3, because the TOP orientation produces the highest electric field strength in the truncated head models due to the largest cross-sectional area in the axial plane of the truncated heads. In LF dosimetry, the skin in the trunk is usually considered as the surrogate region for the consideration of PNS stimulation [32], [33], [49], because of the occurrence of the highest induced electric field strength owing to the large coronal section area. For the simulations using head models in this study, we mainly considered the fields in brain tissues, for which the tissue smoothness is important due to the complex anatomy.

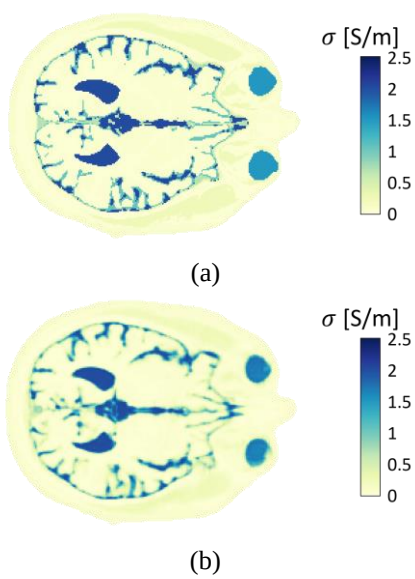

Fig. 2. Conductivity distributions in the axial slice of (a) segmented and (b) learning-based head models for a subject (no. 45) in the Brian Multimodality dataset [47].

C. Electromagnetic Computational Method

At frequencies up to ~10 MHz, the human body is assumed to not perturb the external magnetic field [50]. By ignoring propagation, capacitive and inductive effects, Maxwell's equations were simplified with quasi-static approximation [50]–[52]. Thus, the tissue permitivity can be neglected for the LF electromagnetic computation. The resulting electric scalar potentials for an external magnetic field were computed using the scalar-potential finite difference method:

$$\nabla \cdot [\sigma(-\nabla\varphi - j\omega \boldsymbol{A}_0)] = 0 \tag{1}$$

where $\boldsymbol{A}_0$ and σ denote the magnetic vector potential of the applied magnetic field and the tissue conductivity, respectively. In this study, scalar potential was computed iteratively via the successive-over-relaxation and multigrid methods [53]. When Equation (1) is solved, the induced electric field $\boldsymbol{E}$ is calculated by:

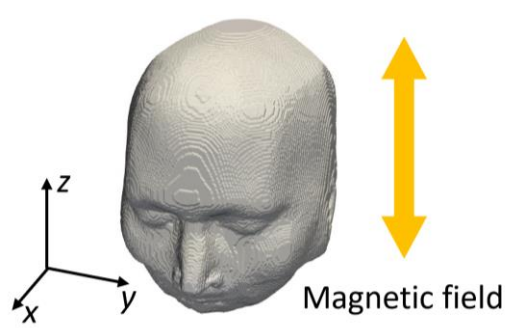

Fig. 3. Exposure scenario where subject head was exposed to uniform magnetic field in TOP direction.

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$$\boldsymbol{E} = -\nabla\varphi - j\omega\boldsymbol{A}_0 \tag{2}$$

D. Post-Processing Methods

The ICNIRP guidelines (2010) used an averaging volume of 2-mm cube together with the 99th percentile for the induced electric field in a targeted tissue as the relevant dosimetric quantity [1]. The IEEE standard prescribed the induced electric field to be averaged over a 5-mm line segment [2]. In our previous study [54], the detailed schemes for both averaging methods were developed. The volume-averaging was defined by Equation (3),

$$E_v(\boldsymbol{r}_c) = \begin{cases} \dfrac{1}{V_1}\left\|\displaystyle\sum_{\boldsymbol{r}\in V_1} v(\boldsymbol{r})\boldsymbol{E}(\boldsymbol{r})\right\|, & \text{if } p < p_{\max} \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

where $\boldsymbol{r}_c$ is the center location of the cube, V_1 is the cubic averaging volume and $v(\boldsymbol{r})$ is the intersected volume of the 2-mm cube with a voxel centered at $\boldsymbol{r}$ ($\boldsymbol{r} \in V_1$). Volume-averaging was applied to all voxels, each centered in their respective cubes.

Line averaging was calculated using the following equation:

$$E_L(\boldsymbol{r}_c) = \begin{cases} \max\left(\dfrac{1}{L_1}\left\|\displaystyle\sum_{\boldsymbol{r}\in L_1} l(\boldsymbol{r})\boldsymbol{E}(\boldsymbol{r})\right\|\right), & \text{if } p < p_{\max} \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

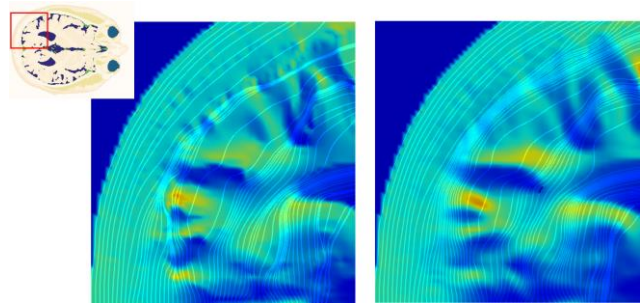
where $l(\boldsymbol{r})$ is the length of the intersected segment of the 5-mm line with the voxel centered at $\boldsymbol{r}$. For both averaging approaches, the maximum permissible percentage of inclusion of other tissues/air, p, was set at 30% (p_{max}), according to our previous study [54]. For the learning-based non-uniform models, tissue labels in the segmented head were adopted for identifying the voxel electric field in the targeted tissue.

After spatial averaging, the percentile values were computed. The 99th percentile values have been introduced as the proper value for internal induced electric field to remove unrealistic computational artifacts [27]. The ICNIRP guideline [1] adopted this. However, it was reported that the 99th percentile value may still underestimate the induced electric field, particularly for non-uniform exposure [36]. Thus, higher percentile values (≥99.9th percentile) were also suggested [35], [36]. In this study, we compared the 99th to 100th percentile values to compare the results between the segmented and non-uniform head models.

III. Results

A. The Induced Electric Field Distributions

Fig. 4 shows the computed induced electric field distributions on the horizontal sections of the segmented and non-uniform head models of a subject (no. 45 in the Brian Multimodality dataset [47]). The induced electric field for both types of models agree with high consistency. It can be observed that the field inside the non-uniform model is more smoothly distributed. Due to the hard tissue boundaries in segmented head models, the induced electric field suffer from the staircasing error.



(a) (b)

Fig. 4. Induced electric field distributions within axial slice of (a) segmented and (b) non-uniform head models for a subject. The streamlines indicate the directions of the electric fields.

B. Percentile Values of Averaged Electric Fields in GM and WM

Fig. 5 and Table II show the percentile values of the cube- and line-averaged electric field in the GM of eight head models. The relative standard deviations (RSDs) for the segmented models were larger than those for the non-uniform models for both cube- and line-averaged values. Moreover, the RSDs were larger for the higher percentile electric fields. The mean or median of the 99th to 99.99th percentile values for both the segmented and non-uniform models had high consistency. In Table II, the relative differences (RDs) between the percentile electric fields obtained using the non-uniform and segmented models were within 9%, with the largest RD occurring for the maximum electric fields (100th percentile value).

Percentile values of the averaged electric fields in the WM are shown in Fig. 6 and Table III. The tendencies are similar to those previously shown for the GM. The RSDs for both models are within approximately 10%, with a maximum value of 12.5% for the 100th percentile value of the cube-averaged electric field for segmented models. For the cube-averaged electric field in the segmented models, the variations were larger for the GM than for the WM. In addition to the segmentation of the complex anatomy, it was also attributed to the high conductivity ratio of CSF to GM than that of GM to WM. Table III shows the differences between the percentile electric fields in WM obtained using the non-uniform and segmented models. The tendency is similar to that in Table II. The RDs were slightly higher than those shown in Table II for GM. The highest RD between a non-uniform and a segmented model was 17.7% for peak cube-averaged field. In Figs. 4 and 5, the RDs and interquartile ranges of the line-averaged electric fields were smaller or narrower than those of the cube-averaged values. This is consistent with our previous findings [54] and is mainly attributed to the larger stencil dimension for the 5-mm line averaging.

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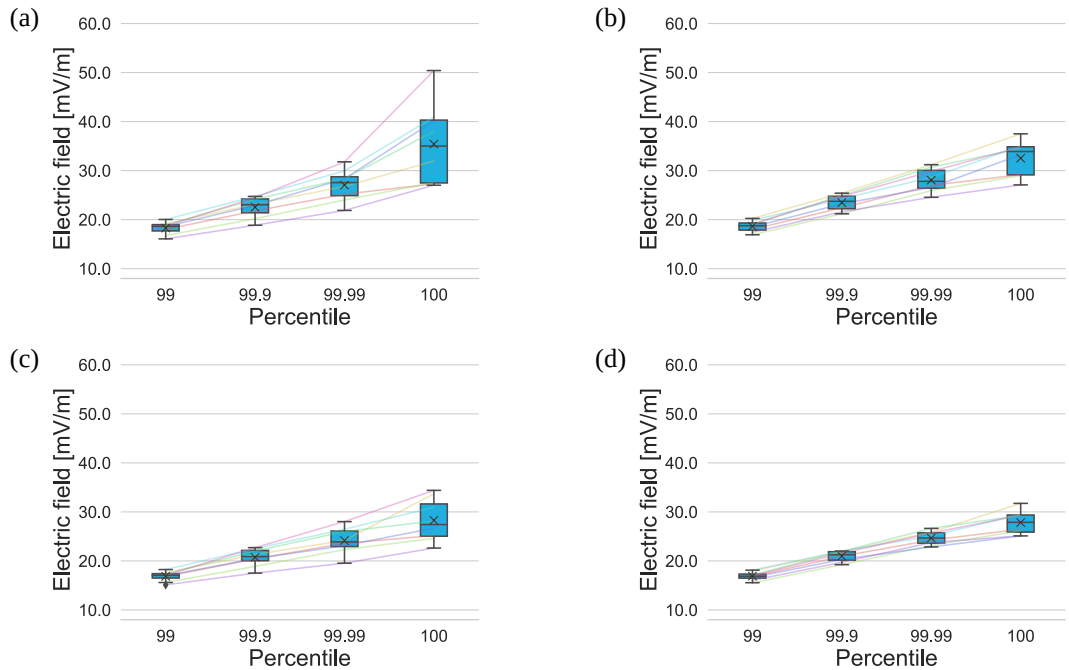

Fig. 5. Percentile values of cube-averaged electric field in GM for (a) segmented model and (b) for non-uniform model and line-averaged electric field in GM for (c) segmented model and (d) for non-uniform model.

TABLE II
MEAN AND MEDIAN PERCENTILE VALUES OF AVERAGED ELECTRIC FIELD IN THE GREY MATTER (GM) OF NON-UNIFORM AND SEGMENTED MODELS.

	%ile	Non-uniform models			Segmented models		
		Mean [mV/m]	Median [mV/m]	RSD [%]	Mean [mV/m](RD%)	Median [mV/m] (RD%)	RSD [%]
Cube-average	99	18.6	18.8	5.8%	18.3 (2.0%)	18.6 (0.7%)	6.6%
	99.9	23.5	23.7	6.4%	22.5 (4.2%)	23.0 (3.1%)	8.8%
	99.99	28.1	27.8	8.0%	27.0 (3.8%)	27.6 (0.9%)	11.2%
	100	32.5	33.9	10.5%	35.4 (-8.1%)	35.0 (-3.2%)	22.2%
Line-average	99	16.9	16.9	4.8%	16.8 (0.3%)	17.1 (-1.0%)	5.7%
	99.9	21.0	21.3	4.8%	20.7 (1.1%)	20.9 (1.9%)	8.2%
	99.99	24.7	24.7	5.2%	24.2 (2.1%)	23.9 (3.3%)	10.3%
	100	27.8	27.9	8.2%	28.3 (-1.6%)	27.4 (1.7%)	14.3%

C. Locations of the Highest Averaged Electric Fields in GM and WM

Figs. 7 and 8 show the distributions of the averaged electric field in GM and WM, respectively. The locations of the top 0.01% of the voxels with high averaged electric fields were denoted by red spheres. These highest electric fields are commonly considered as results of computational artifacts. The locations of the highest electric field were different in the segmented and non-uniform models. Red spheres in the segmented models presented with a more scattered distribution. The locations of the spheres were rather consistent in the non-uniform models of the cube- and line-averaged fields.

IV. DISCUSSION AND CONCLUDING REMARKS

This study assessed the induced electric field in human heads exposed to uniform LF magnetic fields. The electric fields in eight non-uniform head models generated by CondNet learning network were evaluated, and were compared with those obtained using segmented head models with uniform conductivity in each tissue. In general, the electric field distributions produced by the two types of models were consistent. However, there were staircasing artifacts in the head model with uniform conductivity, whereas the non-uniform models can eliminate the staircasing error well because of the smooth transition of the conductivity values at the tissue interface. For non-uniform models, the inter-subject variations of the percentile electric field were also reduced (See Figs. 4 and 5). The reduction was more pronounced for the maximum average electric field. However, potential errors in the maximum value of the electric field cannot be identified, as observed in [34], [37].

The 99th percentile value was introduced as a method to reduce the staircasing error for numerical dosimetry [27]. The

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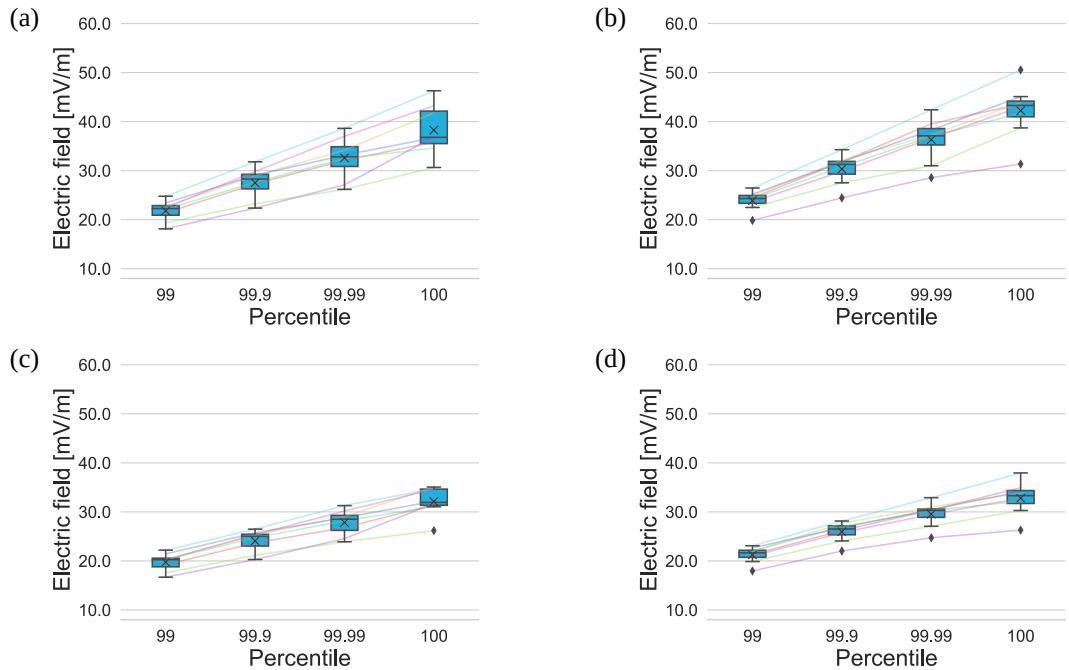

Fig. 6. Percentile values of cube-averaged electric field in WM for (a) segmented model and (b) for non-uniform model and line-averaged electric field in WM for (c) segmented model and (d) for non-uniform model.

TABLE III
MEAN AND MEDIAN PERCENTILE VALUES OF AVERAGED ELECTRIC FIELD IN THE WHITE MATTER (WM) OF NON-UNIFORM AND SEGMENTED MODELS.

	%ile	Non-uniform models			Segmented models		
		Mean [mV/m]	Median [mV/m]	RSD [%]	Mean [mV/m](RD%)	Median [mV/m] (RD%)	RSD [%]
Cube-average	99	23.9	24.3	7.8%	21.8 (9.6%)	22.3 (9.2%)	9.2%
	99.9	30.3	31.3	9.5%	27.5 (10.2%)	28.3 (10.6%)	11.0%
	99.99	36.3	37.1	11.6%	32.6 (11.4%)	32.8 (13.0%)	12.4%
	100	42.2	43.3	12.2%	38.2 (10.5%)	36.8 (17.7%)	12.5%
Line-average	99	21.2	21.6	7.3%	19.7 (7.6%)	20.3 (6.9%)	8.8%
	99.9	26.0	26.5	7.3%	24.1 (7.8%)	25.0 (6.0%)	8.7%
	99.99	29.5	30.3	8.0%	27.8 (6.1%)	28.5 (6.3%)	8.7%
	100	32.8	33.3	9.8%	32.1 (2.1%)	31.9 (4.3%)	8.4%

applicability of the 99th percentile value is case-specific, depending on the model resolution and the number of voxels [17], [29], [55]. For localized exposure, the 99th percentile value may potentially ignore truly high electric field in the target tissue [36], particularly prominent if the voxel number in the tissue is large as more voxels are excluded. Note that the role of the ICNIRP guidelines (2010) is to set the limit or to derive the external and induced field strength rather than to consider the practical scenarios. As shown in Figs. 7 and 8, the highest averaged electric fields are clustered in non-uniform models unlike in segmented models and are generally located around the borders of the section with the largest area, indicating that the staircasing error that appeared at other stepped regions were suppressed to a certain degree. Note that artifacts are difficult to remove completely in any method [56]. However, the proposed non-uniform model mapped directly from the medical images may provide some insights to estimate a more proper dose metric in the exposed tissue.

In previous studies [14], [57], the ratio of the maximum electric field to the 99th percentile value in the brain was generally higher than 2, and it increased with refined model resolution [17]. Comparing the 99th percentile and the maximum value of cube-averaged electric fields in our study, the difference was up to 2.5 in the GM of the segmented model with a resolution of 1 mm and was less than 2 for the smoothed model for GM (Fig. 4). The difference was somewhat suppressed in WM (< 2). One potential reason for this would be the CSF, which has larger conductivity. The difference between the 99th and maximum electric field was somewhat smaller for line-averaged values. This may depend on the averaging algorithm, as discussed in [54].

The computational results in [14] were utilized for the

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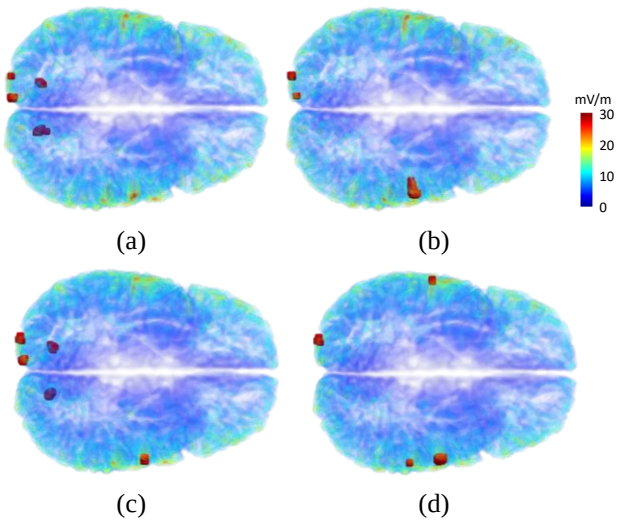

Fig. 7. Distributions of the cube-averaged electric fields in grey matter (GM) for (a) segmented model and (b) non-uniform model and of the line-averaged electric fields in GM for (c) segmented model and (d) non-uniform model. Red spheres indicate the locations of the top 0.01% averaged electric fields.

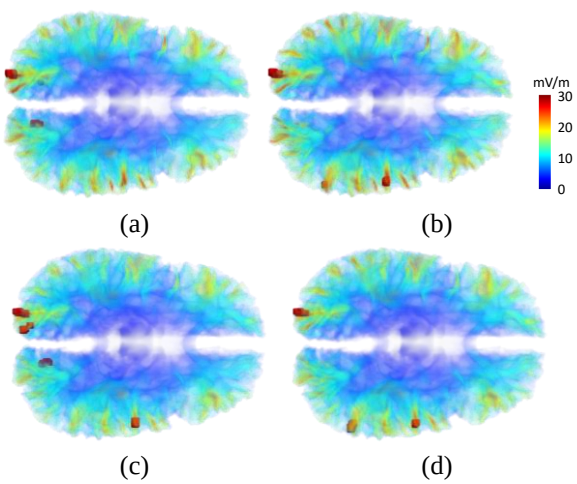

Fig. 8. Distributions of the cube-averaged electric fields in white matter (WM) for (a) segmented model and (b) non-uniform model and of the line-averaged electric fields in WM for (c) segmented model and (d) non-uniform model. Red spheres indicate the locations of the top 0.01% averaged electric fields.

development of the limits in the ICNIRP guidelines 2010. The computed 99th percentile electric fields in the brain were 25.1 mV/m and 22.1 mV/m for the magnetic field in the TOP direction in the NAOMI and NORMAN models, respectively. In [57], the 99th percentile values of the induced electric field were 22.9 mV/m and 20.1 mV/m in the brains of the TARO and HANAKO head models, respectively, for 1 mT magnetic field in the TOP direction at 50 Hz. Our results for TOP magnetic exposure (with a median of 24.3 mV/m and RSD of 7.8% in the non-uniform WM) were close to the previously reported values. In the IEEE standard [2], magnetic induction was computed using homogeneous ellipsoidal models. For the exposure of the ellipsoid to magnetic field at 50 Hz, the induced electric field can be calculated to be about 16.3 mV/m per mT. This value is lower because only homogeneous canonical model was considered [58].

One finding in this study was that even considering the maximum value of electric field which still includes stair-casing error [37], the uncertainty caused by the conductivity assignment was less than 2.5. This value was smaller than 3 defined in the ICNIRP guidelines which consider overall numerical uncertainty. Excluding the maximum value, the numerical uncertainty in different models was the order of 20%. Unlike the international guidelines, our finding cannot be applied directly to the limit because additional computational artifact can be found around the armpit, fingers etc. Nonetheless, the findings caused by the conductivity assignment could be useful for setting the appropriate reduction factor in future revisions of international guidelines and standards. An additional uncertainty factor not considered here was due to the computational code. To resolve this, the intercomparison of computational results is essential as listed in [7] and remains future study.

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